

CONVERSION OF INFIX TO PREFIX:

DATE:27-10-2025

CODE:

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>

char s[50]; int top=-1;

void push(char c){ s[++top]=c; }

char pop(){ return s[top--]; }

int prec(char c){ return (c=='+'||c=='-')?1:(c=='*'||c=='/'||c=='^')?2:0; }

void rev(char *a){
    int n=strlen(a),i=0;
    for(i=0;i<n/2;i++){
        char t=a[i]; a[i]=a[n-i-1]; a[n-i-1]=t;
    }
    for( i=0;i<n;i++){
        if(a[i]=='(') a[i]=')';
        else if(a[i]==')') a[i]='(';
    }
}

void infixToPrefix(char *in){
    char out[50]; int k=0,i=0; rev(in);
    for( i=0;in[i];i++){
```

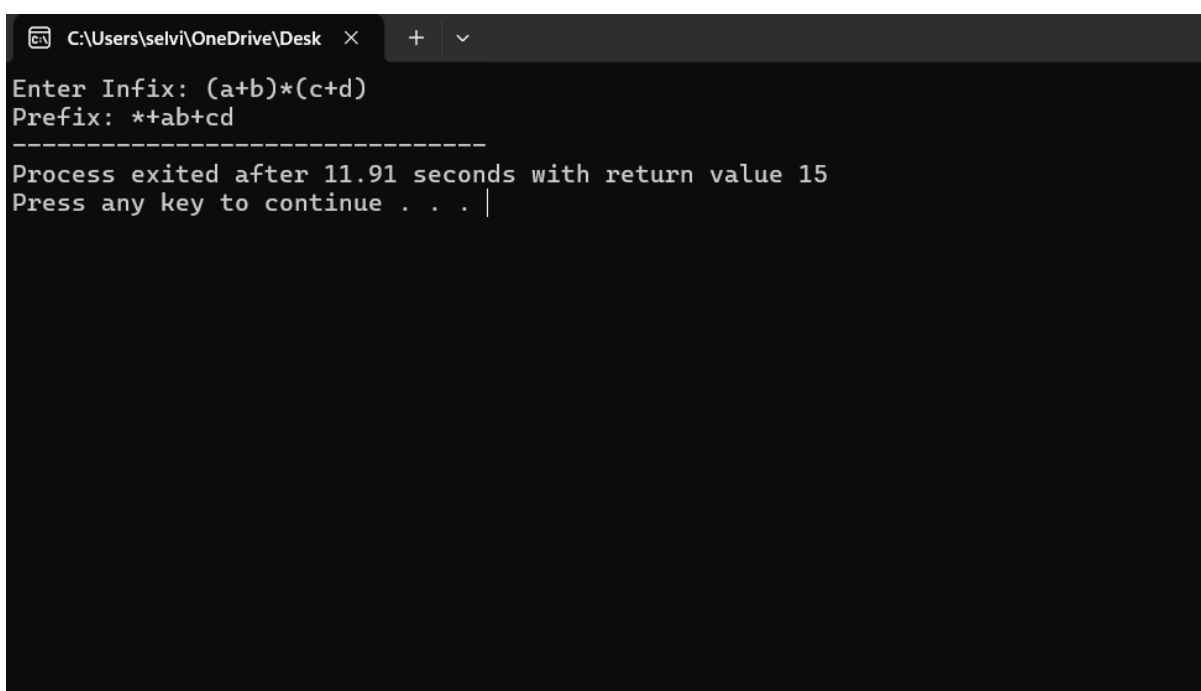
```

    char c=in[i];
    if(isalnum(c)) out[k++]=c;
    else if(c=='(') push(c);
    else if(c==')'){ while(s[top]!='(') out[k++]=pop(); pop(); }
    else{ while(top!=-1 && prec(s[top])>=prec(c)) out[k++]=pop(); push(c); }
}
while(top!=-1) out[k++]=pop(); out[k]='\0'; rev(out);
printf("Prefix: %s",out);
}

int main(){
    char in[50];
    printf("Enter Infix: "); gets(in);
    infixToPrefix(in);
}

```

OUTPUT:



```

C:\Users\sarvi\OneDrive\Desktop >
Enter Infix: (a+b)*(c+d)
Prefix: **ab+cd
-----
Process exited after 11.91 seconds with return value 15
Press any key to continue . . .

```

